# Random Walks: What Does a Blind Walker Learn?

Module 7 · Advanced Network Science
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(school logo, bottom left, as in the CCS deck)

#1 title

## A walker with no map takes random steps. What does it learn about the network?

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#2 spine 1: the question

## One step: pick a neighbour at random

FIGURE

Small 7-node network, hand-drawn. A walker (ring) stands on a node with 4 neighbours. Each of the 4 edges is labelled 1/4 (an amount, not a share). A grey aside: 'no memory of where it came from'.

3

#3 mechanism: the rule

LIVE TOOL

LIVE: random walk simulator, karate club, exactly as used last term (assets/vis/random-walks/index.html, no edits). The lecturer clicks Walk x20, then Run 5,000 steps.

4

#4 live: the simulator (karate club)

## Short walk vs long walk

20 steps: what do you learn?
5,000 steps: what do you learn?

FIGURE

Two small screenshots of the tool: after 20 steps, after 5,000 steps (no answer marks yet)

Take a minute. Tell your neighbour.

5

#5 question to the room (Q1)

## Short: neighbourhood. Long: degree.

FIGURE

LEFT: the tool after 20 steps, nodes shaded by visits: dark nodes cluster around the start node. RIGHT: after 5,000 steps: dark nodes are the hubs (33 and 0). Grey ring call-outs: 'the start', 'hubs'.

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#6 answer to Q1

LIVE TOOL

LIVE: same tool, ?dataFile=two-cliques.json: two 5-cliques joined by a single edge.

7

#7 live: the simulator (two rooms)

## Two rooms, one door

How many steps until the walker crosses to the other room?

A: about 3
B: about 20
C: about 200

FIGURE

The 10-node two-room network, the walker in room 1, the single edge between the rooms marked with a grey arrow

Take a guess.

8

#8 question to the room (Q2)

## A short walk stays in its room

FIGURE

Same network. The walker's path drawn as a brush stroke, coloured by room (room 1 cool, room 2 hot); a time strip under it shows long runs of one colour, about 21 steps each. Grey call-out: 'about 21 steps per visit'.

9

#9 answer to Q2

## Short walks see rooms. Long walks see degree. When does one turn into the other?

10

#10 spine 2: the mystery

## Pen & Paper, Part 1: measure it

10 minutes · stop at the line on the sheet

FIGURE

QR code (large) to the simulator, plus the short link

11

#11 Pen & Paper Part 1: opener

## Part 1: three things to do

1. Sketch: correlation with degree over time
2. Run 20,000 steps. Read the most and least visited node.
3. Compute degree ÷ share, for both

FIGURE

Large timer (10:00) and the QR code

12

#12 Pen & Paper Part 1: instructions

## Correlation climbs, then flattens

FIGURE

Line plot. x: steps (log scale, 10 to 20,000). y: correlation between visit share and degree. Dots at 0.37, 0.53, 0.76, 0.89, 0.96, 0.99, 1.00 (mean over all 34 starts), a pale band for min to max. Grey call-out: 'at 20 steps, it depends on where you start'.

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#13 debrief Q1 (the sketch)

## Degree ÷ share is one number

Karate club: 34 nodes, 78 edges, so 2m = 156

FIGURE

Scatter: x = degree, y = share of visits after 20,000 steps, 34 dots on a straight line through the origin. Node 33 (degree 17, share 0.109) and node 11 (degree 1, share 0.0064) circled, hot. Grey call-outs: '17 ÷ 0.109 = 156', '1 ÷ 0.0064 = 156'.

**Share of time = degree ÷ 2m**

14

#14 debrief Q2 (degree ÷ share)

## Why does the walker forget where it started?

15

#15 spine 3: part question

## The whole walk is one matrix

FIGURE

LEFT (small): the 12-node 'two rooms' network (nodes 0-5 and 6-11, two bridge edges 2-8 and 5-11). RIGHT: its transition matrix P as a heat map, 12 x 12, two dense blocks on the diagonal, thin entries off the blocks. Grey call-out on one row: 'row i = the odds of leaving i, sums to 1'.

16

#16 P (the 'two rooms' network, the motif of the rest o

## One step is one multiplication

$$x(t+1) = x(t)\,P$$

FIGURE

Matrix picture in the M06 style: a wide bar (distribution x(t), 12 cells, shaded) times the square P = a wide bar (x(t+1)). Small network beside it with the walker's probability mass drawn as node shading. Grey call-out: 'x is a row vector: row x column'.

**After $t$ steps: $x(t) = x(0)\,P^t$**

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#17 x P

build 1/3

## Rows of $P^t$ after t steps

FIGURE

Heat map of P^1 (12 x 12), same crop and position as the previous step: two dense diagonal blocks. Callout: 't = 1: stays in its room'

18

#18 P^t as heat maps (1/3)

build 2/3

## Rows of $P^t$ after t steps

FIGURE

Heat map of P^5 (12 x 12), same crop and position as the previous step: the blocks blur. Callout: 't = 5: starting to leak'

19

#19 P^t as heat maps (2/3)

build 3/3

## Rows of $P^t$ after t steps

FIGURE

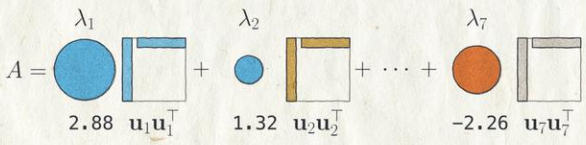
Heat map of P^40 (12 x 12), same crop and position as the previous step: every row identical. Callout: 't = 40: all rows = pi'

**Every row becomes the same: $\pi$**

20

#20 P^t as heat maps (3/3)

## Recall: a matrix = circles and bars




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#21 recap: eigendecomposition (M06, light touch)

## $P$ is a sum of circles and bars too

$$P = \textstyle\sum_k \lambda_k\,\phi_k\,\psi_k^\top$$

FIGURE

Same pictorial grammar as M06: circle (eigenvalue, size = |lambda|), tall bar phi_k (right eigenvector), wide bar psi_k^T (left eigenvector); k = 1 blue, k = 2 gold, the rest pale; shown as terms 1, 2, ..., 12. A grey aside: 'for P, the tall and wide bars are two different vectors (each is the other rescaled by degree)'.

**Powers only touch the circles**

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#22 P as a sum of circles and bars

build 1/3

## Each step shrinks the circles

FIGURE

P^1 drawn as a sum of the same terms (same canvas as the previous step); circle areas = lambda^1: circles at 1, 0.89, 0, 0, 0, -0.09, -0.2, -0.2, -0.27, -0.33, -0.38, -0.42. First circle always full size, blue. Callout: 'one step'

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#23 P^t as circles (1/3)

build 2/3

## Each step shrinks the circles

FIGURE

P^5 drawn as a sum of the same terms (same canvas as the previous step); circle areas = lambda^5: circle 2 at 0.55, the rest almost gone. First circle always full size, blue. Callout: 'lambda_2^5 = 0.55'

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#24 P^t as circles (2/3)

build 3/3

## Each step shrinks the circles

FIGURE

P^40 drawn as a sum of the same terms (same canvas as the previous step); circle areas = lambda^40: circle 2 at 0.01; circle 1 untouched. First circle always full size, blue. Callout: 'only the circle at 1 is left: every row is pi'

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#25 P^t as circles (3/3)

## Which circle dies last?

The 12 circles for $t = 1$.

After many steps, which one is still there?

FIGURE

The 12 eigenvalue circles in a row, ordered by value; no mark yet

Take 20 seconds.

26

#26 Q3 (which circle dies last)

## The second circle dies last

$\lambda_2 = 0.89$

$0.89^t$: 0.55 at $t = 5$, 0.31 at $t = 10$, 0.01 at $t = 40$

FIGURE

Same 12 circles, now drawn at t = 10 sizes; circle 2 hot, the rest pale

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#27 A3

## Forgetting is a distance that shrinks

FIGURE

Five small bar charts in a row: the walker's distribution x(t) on the 12 nodes for t = 0, 1, 3, 10, 40, starting at node 0. A flat reference line pi on each chart (pi = degree / 60). Under each: 'distance to pi' as a number (1-norm), shrinking to about 0.

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#28 distance to pi (small multiples)

build 1/3

## Mixing time: every start forgotten

FIGURE

Curves: distance to pi against t, one thin grey line per starting node (12), and a black line for the worst start. Callout: 'where you start matters at first'.

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#29 mixing time diagram (1/3)

build 2/3

## Mixing time: every start forgotten

FIGURE

Same plot plus a horizontal line at epsilon. Callout: 'how close is close enough?'

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#30 mixing time diagram (2/3)

build 3/3

## Mixing time: every start forgotten

$t_{\text{mix}} = \min\{ t : \max_{x(0)} \|x(t) - \pi\|_1 \leq \epsilon \}$

FIGURE

Same plot; the point where the worst-start line crosses epsilon marked hot, with a vertical line down to the axis labelled 'mixing time'.

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#31 mixing time diagram (3/3)

## Q: how long to forget on a 10-clique?

10 nodes, every pair connected (45 edges)

FIGURE

A 10-clique, hand-drawn, nodes in a ring; the walker on node 0

Take a guess: 1 step, 5 steps, 50 steps.

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#32 Q4 (clique)

## A: one step is almost enough

10 nodes, every pair connected (45 edges)

FIGURE

Same clique, same position. Distance-to-pi curve as an inset: below epsilon at t = 1

$\lambda_2 = -1/9$: the walker forgets in about 1 step

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#33 A4

## Q: and with a single bridge?

Two 5-cliques joined by one edge (21 edges)

FIGURE

The two-clique network (the tool's two-cliques.json), the walker in room 1

Same, 5x, or 15x slower than the clique?

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#34 Q5 (two cliques)

## A: fifteen times slower

Two 5-cliques joined by one edge (21 edges)

FIGURE

Same network. Inset: distance-to-pi curve with the clique's curve in pale for comparison

$\lambda_2 = 0.93$: about 14 steps to forget

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#35 A5

## The slowest circle sets the clock

$$\tau = \frac{1}{1 - \lambda_2}$$

FIGURE

A horizontal gauge for lambda_2 from -0.2 to 1 with markers: 10-clique (-0.11, tau 0.9), karate club (0.87, tau 7.6), two rooms (0.89, tau 9), two 5-cliques + bridge (0.93, tau 14), ring of 1,000 (0.99998, tau 50,000). Past 0.99 the gauge is shaded hot.

Small gap, long memory

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#36 spectral gap (gauge)

## Mixing time is a few relaxation times

OPTIONAL

FIGURE

The formula t_mix < tau log(1/(epsilon pi_min)) on the left; on the right the 12-node distance curve with tau marked and 2 tau, 3 tau as ticks.

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#37 (optional) the bound

## One eigenvalue is always 1. None is larger.

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#38 spine 4: the fact

## Why 1? Every row of $P$ sums to 1

FIGURE

Picture: the square P times a tall bar of all ones (cells '1') = the same tall bar. One row of P highlighted; its cells sum to the first cell of the result (= 1). Callout: 'P 1 = 1'.

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#39 lambda = 1 exists (P 1 = 1)

## Why not more? An average can't grow

FIGURE

Two frames on a 7-node network: numbers on the nodes (largest = 9); one step replaces every number by the average of its neighbours; the largest number can only stay or fall. Frame 2 is the same drawing with the new numbers; the largest is now 6.5.

40

#40 no eigenvalue above 1

## The circle at 1 is π

The stationary distribution: $P^t \to \mathbf{1}\,\pi$

FIGURE

The first term alone: a tall bar of ones times a wide bar pi = a matrix whose every row is pi (the last heat map of the previous build, at the same position). Beside it the simulator's bars with the orange k/2m ticks (thumbnail).

$\pi_i = k_i/2m$: degree, and nothing else

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#41 lambda = 1 is pi (ties to the tool)

## Pen & Paper, Parts 2 and 3: expand it

20 minutes

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#42 Pen & Paper Parts 2-3: opener

## Parts 2 and 3: two things to do

2. Four nodes: write $x(t)$ with the eigenvalues, check $t = 2$
3. Twelve nodes: run one cell, plot the second eigenvector

FIGURE

Left: the 4-node "diamond" network of Part 2. Right: QR code to the one-cell notebook of Part 3

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#43 Pen & Paper Parts 2-3: instructions

## $x(t)$ = π + shrinking pieces

Four nodes; eigenvalues $1,\ 0,\ -\frac{1}{3},\ -\frac{2}{3}$

FIGURE

The 4-node diamond network (degrees 2, 2, 3, 3) on the left. Right: bar charts of x(t) for t = 0, 1, 2, 5, 10 converging to pi = (0.2, 0.2, 0.3, 0.3). Under them the expansion: pi - 0.5 (0)^t psi_2 + 0 psi_3 + 0.3 (-2/3)^t psi_4, with the t = 2 line checked against x(2) = (1/3, 1/3, 1/6, 1/6).

Only the term with $\lambda = 1$ stays

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#44 debrief Part 2

## The second eigenvector splits the rooms

FIGURE

Point plot: x = node index 0 to 11, y = second eigenvector of P (scaled to max 1). Nodes 0-5 on top (about +1; the bridge nodes 2 and 5 at about +0.74), nodes 6-11 on the bottom (about -1; 8 and 11 at -0.74). Room 1 cool, room 2 hot. A dashed line at 0. Callout: 'the sign tells the room'.

45

#45 debrief Part 3 (second eigenvector)

## Early on, the community is what is left

Walker starts at node 0

FIGURE

Line plot: x = steps (0 to 40), y = probability of being in room 1: 1.00, 1.00, 0.92, 0.88, 0.80 (t = 5), 0.67 (t = 10), 0.55 (t = 20), 0.505 (t = 40). A dashed curve 0.5 + 0.5 x 0.889^t over it. Callout at t about 3: 'the other eleven circles are gone'.

Rooms fade like $\lambda_2^t$. Degree is what remains.

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#46 debrief Part 3 (what dominates early)

FIGURE

Summary, three rows, each with a thumbnail of its figure: (1) the circle at 1 = degree (thumb: scatter, slide 14). (2) The second circle = the rooms, and its size sets the mixing time (thumb: gauge). (3) Short walks see rooms; long walks see degree (thumb: the room-probability curve).

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#47 summary (claims with thumbnails)

## Next: how long does the walker take to get there?

FIGURE

The opening picture again: the karate club, nodes shaded by long-run visits, with two nodes marked A and B

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#48 close: back to the opening picture, then a question